

Decision Support System

DSS = Decision Support System



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OLAP AND DATA CUBES

A simple guide to understanding how businesses analyze data from many angles



OLAP Concept



Data Cube



OLAP Operations



Real Uses

What is OLAP?



In simple words

OLAP stands for Online Analytical Processing. It is a way to look at data from many different angles at the same time, so people can find patterns and make better decisions.

- ✦ Instead of one flat table, OLAP organizes data by categories, like Product, Time and Region
- ✦ It lets you quickly summarize big numbers, or zoom into small details, with a few clicks
- ✦ It is built for asking questions like: "How did sales grow by region last quarter?"
- ✦ OLAP tools power the reports and dashboards used in business analytics



A quick example

A shop wants to know:

"Which product sold best in which city, in which month?"



Product: Shoes, Bags, Watches...



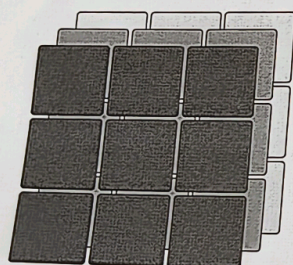
City: Chennai, Delhi, Mumbai...



Month: Jan, Feb, Mar...

OLAP lets the shop combine all three at once — and switch views instantly, without writing new queries every time.

Data Cube Structure



Product

Region

Time

Each small box is one "cell" — one exact fact, like sales of shoes in Chennai in March.



Dimensions

The different angles you can look at data from — like Product, Time, or Region. Think of them as the sides of the cube.



Measures

The actual numbers you care about — like Sales Amount, Quantity Sold, or Profit.



Cells

Each small box inside the cube. It holds one measure value for one combination of dimensions.



Hierarchies

Levels inside a dimension: like Year > Quarter > Month, or Country > State > City.

OLAP Operations

Four simple actions used to explore a data cube



ZOOM OUT

Roll-up

Combine small details into a bigger summary. Example: adding up daily sales to see the total for the whole year.



ZOOM IN

Drill-down

Break a summary into smaller details. Example: opening up yearly sales to see sales for each month.



ONE LAYER

Slice

Pick just one value from a dimension and view that single layer. Example: looking only at sales from the year 2025.



SMALLER CUBE

Dice

Pick a few values from two or more dimensions to make a smaller cube. Example: sales of Shoes and Bags in Chennai and Delhi.

Applications in Analytics

Where OLAP and data cubes are used in the real world



Retail & Sales

See which products sell best, in which store, in which season



Finance

Track budgets, expenses and profits across departments and time



Marketing

Study campaign results across regions, age groups and channels



Business Reports

Build dashboards that managers use to make fast decisions



Inventory & Supply

Watch stock levels by warehouse, product and time period



Healthcare

Analyze patient data by hospital, treatment type and outcome